

How AI Chatbot Response Style Affects Cognitive Load and Performance in Educational Tasks

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Abstract. As conversational AI agents are increasingly integrated into educational settings, understanding their cognitive impact on learners is essential. This study quantifies the cognitive load experienced by students solving GRE-style verbal reasoning problems with and without generative AI chatbot assistance. Using a within-subjects Wizard-of-Oz design, 31 university students completed equivalent verbal tasks under four controlled response conditions—Standard, Lengthy, Unstructured, and Ambiguous. We combined eye tracking metrics (pupil diameter and fixation duration) with subjective workload ratings to capture both autonomic and perceptual dimensions of mental effort. Results demonstrate that AI assistance improves overall task accuracy (from 30.6% to 74.0%) but that response quality critically modulates cognitive load. Specifically, Lengthy, Unstructured, and Ambiguous chatbot outputs elicited subjective scores, whereas concise, Structured (Standard) responses minimized both physiological arousal and perceived effort. These findings offer concrete design guidelines for chatbots, highlighting the value of clear, structured, and succinct responses to maximize learner success and minimize unnecessary effort.

Keywords: AI Chatbots · Cognitive Load · Conversational User Interface · Eye-tracking · Mixed-methods Research

1 Introduction

Generative AI chatbots have emerged as powerful tools in educational settings, transforming the ways in which learners engage with information and interact during their learning processes. Their intuitive interfaces, reminiscent of search bars or messaging apps, contribute to a positive perception among students, who often find them easy to use and helpful in various learning contexts (Feng et al., 2024). These AI-powered tools are increasingly being utilized by students to quickly grasp complex topics and receive supplementary learning assistance (Holmes et al., 2019). Despite their potential benefits, the user experience with Generative AI chatbots can be significantly affected by high cognitive load, leading to confusion, frustration, and a reluctance to engage with these tools.

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Although prior research has explored the cognitive effects of multimedia learning environments (Mayer, 2005), we still lack a clear understanding of how conversational user interfaces (CUIs) such as ChatGPT impact users' cognitive load. According to cognitive load theory (Sweller, 1988), human working memory is limited, so it is essential to design interfaces that minimize unnecessary mental processing. Building on Cognitive Load Theory's principles, we turn to AI specific factors. Generative AI chatbots draw on vast datasets of text from diverse sources (Dergaa et al., 2024), which gives rise to a variety of response styles shaped by differing writing conventions, subject domains, levels of formality, and cultural contexts.

In human–computer interaction literature, it is well established that the total cognitive load directly affects how easily users locate information and complete tasks (Whitenton, 2013). Consequently, this makes response type variation a critical design consideration. Cognitive load measurement methods can be highly effective in assessing the mental effort required to complete tasks, providing empirical justification for measuring these effects (Darejeh et al., 2024). Cognitive overload is closely associated with task performance and can negatively impact both task completion times and overall efficacy when mental resources are exceeded (Galant-Gołębiewska et al., 2020; Biondi et al., 2020). Therefore, quantifying cognitive overload provides an opportunity to identify and refine specific elements that contribute to increased cognitive demand for the user.

Generative AI chatbots introduce several characteristics that can inadvertently raise users' cognitive load. First, when responses are overly long or information-dense, they risk overwhelming users limited working memory capacity (Seran et al., 2025). Processing and retaining large volumes of text in one try forces users to juggle multiple pieces of information simultaneously, increasing intrinsic cognitive load. Second, ambiguous or inconsistent responses compel users to question and verify what the chatbot has provided, adding an extraneous layer of mental effort. Rather than accepting recommendations at face value, users must engage in additional interpretation and fact-checking; an extra cognitive step that can lead to errors in task performance (Zhai et al., 2024; Chandler & Sweller, 1991). Ambiguous outputs are inherently harder to understand, further burdening users' mental resources.

Third, poorly structured or unorganized replies exacerbate germane cognitive load by forcing users to mentally reorganize information into a coherent format. Although well-designed AI systems can offload routine inquiries and streamline support (Gerlich, 2025), disordered outputs have the opposite effect, making it harder for users to extract key points and follow logical flows. Building on these insights, this study examines how four distinct chatbot response styles—Standard, Lengthy, Unstructured, and Ambiguous—affect physiological indicators of mental effort (RQ1). We also seek to determine which style best balances task performance with minimal cognitive load, thereby guiding the design of conversational AI systems that support effective, cognitively sustainable user experiences (RQ2).

Prior studies have found a relationship between eye tracking data and human mental or emotional states such as stress and cognitive workload. For example, Torres-Salomao et al. (2015) found pupil diameter to be a viable indicator of incremental stress when participants were exposed to math problems at varying levels of difficulty. A recent systematic review of literature from 2010 to 2020 of eye tracking measures used to

assess cognitive workload in surgery found that pupil responses, gaze patterns, and blinks were often associated with workload and stress (Tolvanen et al., 2022). Studies have shown that eye tracking measures are reliable and effective at indicating human psychophysiological responses, such as cognitive workload and stress, during various tasks.

Therefore, the primary objective of this study is to quantify the cognitive load experienced by users when solving verbal reasoning problems with and without assistance from a generative AI chatbot. To achieve this, we employ eye tracking metrics to capture autonomic and ocular indicators of mental effort. By standardizing the AI interaction through a Wizard-of-Oz style chatbot, we ensure that variability in chatbot responses can be systematically manipulated and its effects on cognitive load isolated. Specifically, the study aims to:

- Compare baseline cognitive load between unassisted problem-solving and AI-assisted problem-solving on GRE-style verbal questions of equivalent difficulty.
- Examine the influence of AI response types (Standard, Lengthy, Unstructured, or Ambiguous) on physiological indicators of cognitive load.
- Assess the interaction between AI response types and task performance, evaluating whether certain response styles facilitate or hinder both accuracy and mental effort.

2 Methodology

The experiment was designed to investigate how different styles of AI chatbot responses (e.g., length, ambiguity, unstructured, and standard) influence cognitive load during problem-solving tasks. To assess cognitive load, the study employed a mixed-methods approach, integrating physiological measurements (via eye-tracking) with subjective evaluations collected through post-task surveys. 31 graduate students from Purdue University, aged 18–28, participated in the study. They were recruited from two separate courses and received extra course credit as an incentive for their involvement.

The study was structured into two primary phases designed to evaluate participants' cognitive load and task performance during GRE-style verbal reasoning tasks, both with and without AI assistance.

1. Baseline Phase (No AI Assistance): Participants initially completed four medium-difficulty GRE verbal reasoning questions without any AI support. This phase was designed to establish baseline measures of cognitive load and task performance in the absence of external assistance.
2. AI Interaction Phase: In the second phase, participants completed sixteen additional GRE verbal reasoning questions with AI support. The difficulty level of the questions remained consistent (medium) across all tasks. AI responses were delivered using a Wizard-of-Oz methodology, wherein researchers provided pre-scripted answers representing four distinct response types:

- Standard Response: Concise bullet-point answers enhanced with emojis for clarity and engagement.
- Long Explanatory Answer: Detailed, paragraph-based explanations spanning four to five paragraphs.

- Unstructured Bottom-Up Answer: A progressive list of possible answers, ordered from least to most relevant.
- Ambiguous Answer: Vague and indirect responses lacking definitive guidance.

This within-subject design ensured that each participant was exposed to all four AI response types. A fixed order was used across participants to prevent carryover effects between varying chatbot conditions from contaminating performance or perception within each response type block. This approach prioritized internal consistency of condition-specific responses and minimized fatigue or confusion from frequent switching; however, it may introduce order effects, which are addressed in the limitations. During the experiment, participants wore Tobii Glasses 3 (Fig. 1) to collect eye-tracking data. The eye-tracking system was used to monitor gaze patterns, fixation durations, and pupil dilation, providing continuous, non-intrusive physiological indicators of cognitive load during task performance.



Fig. 1. The Tobii eye-tracking device recorded gaze patterns, fixation durations, and pupil dilation in real time, providing physiological indicators of cognitive load.

Upon completing the experimental tasks, participants filled out a self-report survey designed to assess subjective cognitive workload. The survey included items evaluating mental effort, frustration, and the perceived clarity of the AI-generated responses. Table 1 summarizes the survey questions.

The experiment was conducted in a controlled environment with minimal external distractions (Fig. 2). Upon arrival, participants received a detailed explanation of the study and provided informed consent. The eye-tracking glasses were then calibrated for each participant. The session began with the baseline phase (no AI assistance), where participants completed four GRE-style verbal reasoning questions. This was followed by the AI interaction phase, in which participants worked through sixteen additional questions under the four scripted AI response conditions.

Pupil diameter data were collected for both the left and right eyes. The mean pupil diameter was computed for each eye and averaged to yield an overall value for each participant under each condition. Given that pupil dilation is a well-established measure

Table 1. Subjective Questionnaire

Survey Domain	Question Items
Demographics	– Age – Gender (Male, Female, prefer not to say) – Are you a native English speaker? (Yes/No)
Mental Effort and Cognitive Load	– Mental effort required (1–10 scale) –Was the task more mentally demanding than expected? – Effort to understand AI responses – Feeling overwhelmed by AI information – Frustration with AI interface
Clarity and Structure of AI Responses	– Clarity and conciseness of AI information – Difficulty identifying important information
AI Usability and Interaction	– Ease of interaction with AI assistant – Noticing inconsistencies in AI responses – Impact of inconsistencies on learning – Impact of inconsistencies on trust in AI
Physical Discomfort	– Physical discomfort while using AI interface (Yes/No)
Perceived Learning Support	– AI helped achieve learning objectives – Emoji use improved comprehension
Factors Affecting Comprehension	– Responses too long – Responses unstructured – Responses ambiguous – Responses inconsistent

for cognitive workload, these data were analyzed in conjunction with the subjective survey responses. A correlational analysis was performed to assess the relationship between physiological and self-reported workload, as suggested in previous literature (Torres-Salomao et al., 2015).

Additionally, fixation duration was extracted from the eye-tracking recordings. Fixations, defined as pauses of the eye on a single location, serve as an indicator of cognitive processing (Liang et al., 2021). These durations were analyzed across the different response types.

The formal null and alternative hypotheses for these comparisons can be summarized by the following:

- **Null Hypothesis (H0):** There is no statistically significant difference in eye-tracking metrics or subjective survey responses across the different AI response types, indicating no variation in cognitive workload.
- **Alternative Hypothesis (HA):** The type of AI response significantly influences average pupil diameter, mean fixation duration, and subjective workload scores, indicating differential levels of cognitive workload.

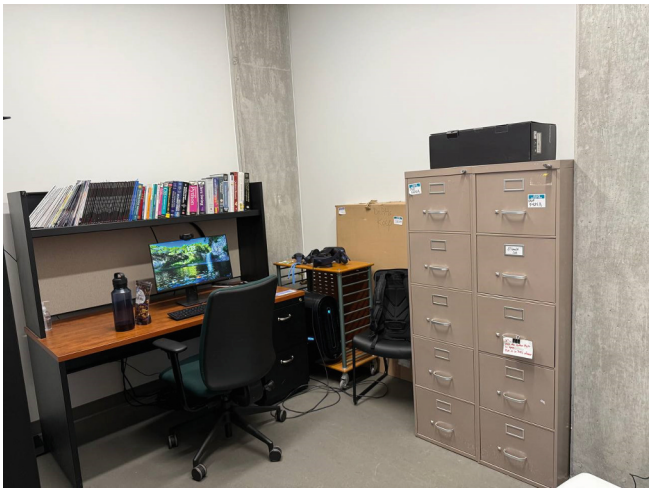


Fig. 2. Experiment Room.

2.1 Analysis Methodology

Physiological Data Analysis. To facilitate data extraction, the timestamps corresponding to each AI response type were manually labeled in the video recordings for each participant. This labeling enabled segmentation of the eye-tracking data according to the response condition. Key physiological indicators of cognitive load (pupil diameter, fixation duration, and gaze patterns) were extracted for each segment corresponding to the different response types.

Subjective Data Analysis. Participants' subjective experiences were evaluated using responses from the post-experiment survey. For each AI response type, average scores were calculated across key dimensions: mental effort, frustration, and clarity. To examine the relationship between self-reported mental effort and task performance accuracy, a Spearman's rank correlation analysis was conducted. In addition, a repeated-measures ANOVA was performed to assess the effect of AI response type on participants' task accuracy, accounting for within-subject variability.

3 Results

3.1 Physiological Measures Analysis

Pupil Dilation. The results for the mean pupil diameter for baseline condition (task without AI assistance) and the different response types (with AI assistance) are shown below with standard deviations in Table 2. The mean pupil diameter in the baseline condition for all participants is 3.44 ± 0.52 which is slightly higher than the mean pupil diameter in all other AI assisted conditions. The reduction in pupil diameter between baseline (no AI) and AI assistance suggests lower cognitive load and arousal with AI support.

Table 2. Mean Pupil diameter by Condition

Task	Mean Pupil diameter (mm)	t-value	p-value
Baseline	3.44 ± 0.52	28.35	0
Standard	3.41 ± 0.55	1.99	0.050
Long	3.31 ± 0.54	−4.58	0.000
Unstructured	3.36 ± 0.55	−1.23	0.220
Ambiguous	3.38 ± 0.56	−0.03	0.973

A linear mixed effects model was fitted to examine how AI generated response type influenced mean pupil diameter. The model's intercept ($\beta = 3.382$, SE = 0.119, $p < 0.001$) represents the average pupil diameter in the baseline condition (no AI assistance). The "Long" response type produced a significant decrease in pupil diameter compared to baseline ($\beta = -0.072$ mm, SE = 0.016, $p < 0.001$), suggesting reduced cognitive load or arousal when participants processed lengthier AI answers. The "Perfect" response type showed a marginally significant increase in pupil diameter ($\beta = 0.031$ mm, SE = 0.016, $p = 0.050$), indicating a trend toward higher engagement with the most coherent AI responses. Neither the "Unstructured" ($\beta = -0.019$ mm, $p = 0.220$) nor the "Ambiguous" ($\beta = -0.001$ mm, $p = 0.973$) response types differed significantly from baseline.

These findings indicate that the form and clarity of AI-generated responses significantly modulate physiological indicators of cognitive workload. Notably, "Long" responses were associated with reduced pupil diameter, suggesting lower mental effort or increased cognitive ease. In contrast, "Standard" responses elicited a marginally increased pupil size, reflecting heightened engagement or deeper processing. The lack of significant changes for "Unstructured" and "Ambiguous" responses further suggests that poorly framed AI outputs neither alleviate nor meaningfully stimulate cognitive processing.

Fixation Duration. The results for the mean fixation duration (ms) for baseline condition (task without AI assistance) and the different response types (with AI assistance) are shown below with standard deviations in Table 3. The increase in fixation duration between baseline (no AI) and AI assistance suggests lower cognitive load and arousal with AI support.

To examine the impact of AI-generated response types on cognitive workload as reflected in eye movement, a linear mixed effects model was conducted. The model revealed a significant main effect of response type on fixation duration ($p < 0.001$), indicating that different types of AI-generated responses were associated with differing levels of visual attention during question answering. Compared to the baseline condition, the "Standard" response type significantly increased fixation duration ($\beta = 75.18$, $p < 0.001$), suggesting a higher level of sustained visual processing. In contrast, the "Long" response type led to a significant reduction in fixation duration ($\beta = -45.70$, $p < 0.001$), while the "Unstructured" response type also showed a significant negative

Table 3. Mean Fixation duration by Condition

Task	Mean Fixation duration (ms)	t-value	p-value
Baseline	492.56 ± 153.01	1.62	0.109
Standard	549.31 ± 207.83	6.16	0
Long	428.43 ± 149.80	−4.02	0
Unstructured	448.07 ± 154.59	−2.29	0.024
Ambiguous	452.28 ± 161.81	−1.92	0.058

effect ($\beta = -26.06$, $p = 0.024$). The "Ambiguous" response type exhibited a marginally non-significant decrease ($\beta = -21.85$, $p = 0.058$), suggesting a possible trend toward reduced cognitive engagement. These results indicate that the structure and quality of AI-generated answers influence participants' eye movement behavior, with more coherent or optimal answers (Standard response) eliciting more sustained attention.

These results indicate that the structure and quality of AI-generated answers influence participants' eye movement behavior, as measured by fixation duration. In particular, more coherent and well-structured responses (e.g., Standard) elicited longer fixation durations, suggesting greater cognitive engagement and sustained visual attention. In contrast, less structured or overly lengthy responses appeared to reduce attention, either by facilitating easier comprehension (as in the Long condition) or by diminishing user engagement (as in the Unstructured and Ambiguous conditions).

3.2 Subjective Measures Analysis

Overall Accuracy. Participants' mean accuracy in the no-AI condition was $M = 0.306 \pm 0.22$, whereas in the AI-assisted condition it increased to $M = 0.74 \pm 0.17$. A paired-samples t-test confirmed that this difference was highly significant, $t(df) = -9.004$, $p < .001$, indicating that AI assistance yielded superior performance on the GRE questions.

Subjective Workload (Post-Survey). Descriptive statistics for each post-survey dimension are presented in Table 4. Accuracy was highest for "Standard" AI responses and lowest for Ambiguous and Unstructured responses (Fig. 3).

Table 4. Post-survey dimension scores.

Dimension	Avg Score (out of 5)
Mental Effort	2.31
Frustration	3
Clarity	2.209

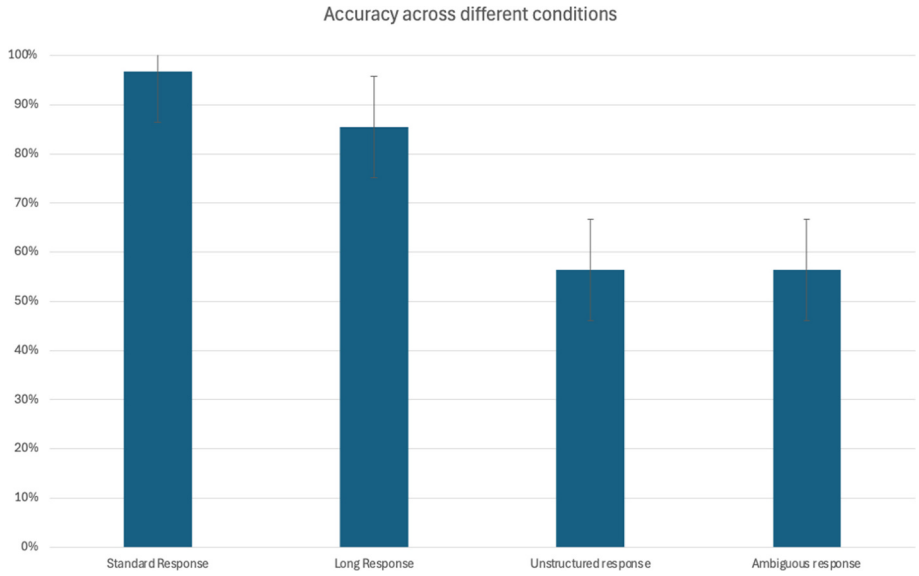


Fig. 3. Accuracy across different conditions.

Correlation Between Accuracy and Mental Effort. A Spearman's rank correlation revealed a significant inverse relationship between AI-assisted accuracy and self-reported mental effort, $\rho = -0.52$, $p = .0025$. Participants who achieved lower accuracy under AI assistance tended to report higher levels of mental effort.

Effect of AI Response Type on Accuracy. A one-way repeated-measures ANOVA examined accuracy across the four AI response categories. There was a significant main effect of response type on accuracy, $F(3, 30) = 3.39$, $p < .001$. Bonferroni-corrected post-hoc comparisons ($\alpha = .0083$) showed that:

- Standard responses produced the highest accuracy (all $p < .001$ vs. other types).
- Lengthy responses yielded intermediate accuracy ($p < .001$ vs. Unstructured and Ambiguous).
- No significant difference emerged between Unstructured and Ambiguous responses ($p = 1.00$).

Native vs. Non-Native English Speakers. Mann–Whitney U tests compared native and non-native speakers on (a) improvement in accuracy from no-AI to AI conditions and (b) mental effort. There was no significant group difference in accuracy improvement (-0.45, 0.649), but non-native speakers reported significantly greater mental effort than native speakers ($U = 0.9$, $p = 0.036$), suggesting differential cognitive demands across language backgrounds.

Participants also reported a moderate level of agreement regarding the educational value of the AI assistant. When asked whether the AI agent helped them achieve the learning objectives, the average response was 3.10 on a 5-point Likert scale (1 = Strongly

Disagree, 5 = Strongly Agree). Interestingly, participants responded more positively to the use of emojis in AI responses. The average agreement scores for the statement "Did the use of emoji help better understand the answers?" was 4.61, indicating that the inclusion of visual cues significantly supported comprehension.

4 Discussion and Limitations

This study provides valuable insights into the cognitive impact of generative AI assistance during complex verbal reasoning tasks; however, several limitations should be noted. While longer fixations are sometimes associated with increased cognitive processing or higher mental effort, in this study they appear to reflect sustained visual attention and engagement, particularly in response to clear and structured AI outputs. This is supported by the fact that the "Standard" condition, which had the highest fixation durations, also produced the highest accuracy and lowest subjective effort. In contrast, shorter fixations observed in the "Lengthy", "Unstructured", and "Ambiguous" conditions may reflect either cognitive disengagement or reduced attentional stability due to information overload or confusion. Thus, fixation duration in this context is best interpreted as a proxy for meaningful engagement rather than raw cognitive load, aligning with prior findings that associate gaze behavior with information relevance and processing fluency (Holmqvist et al., 2011).

First, the study was conducted with a limited and homogeneous sample, which may constrain the generalizability of the findings. Participants' prior familiarity with AI systems or differential cognitive abilities were not controlled, potentially introducing individual variability in responses to AI-assisted conditions; Second, the experimental tasks were administered in a controlled laboratory environment. While this setup allowed for rigorous collection of physiological data, it does not fully replicate the complexity and variability of real-world educational or professional settings. External factors such as environmental distractions, time pressure, and emotional stress (which often influence cognitive load) were minimized and may therefore reduce ecological validity; Third, the AI responses used in the study were static and pre-defined across all participants. In practice, generative AI systems are capable of dynamically adapting to users' behavior and queries. The use of non-interactive responses may limit the ability to fully assess the cognitive benefits or drawbacks of real-time, personalized AI support; Fourth, the interpretation of physiological measures such as pupil dilation and fixation duration, while grounded in prior literature, is inherently multifactorial. Although lighting conditions were controlled and tasks were counterbalanced, these metrics can still be influenced by variables such as fatigue, emotional arousal, or individual differences in oculomotor behavior, which were not fully accounted for; Fifth, the study focused exclusively on GRE-style verbal reasoning tasks. It remains unclear whether the observed cognitive effects of different AI communication styles would generalize to other domains, such as mathematical problem-solving, collaborative decision-making, or clinical reasoning; Sixth, the study captured only the immediate cognitive responses during task execution. Longitudinal outcomes, such as retention, transfer of knowledge, or changes in users' trust and reliance on AI systems over time, were not assessed and warrant further investigation; Lastly, the AI response types were presented in a fixed order for all

participants. While this helped preserve the integrity and coherence of each response condition and avoided confusion from rapidly changing chatbot styles, it does introduce potential order effects such as fatigue, learning, or habituation. These effects could have influenced participants' performance or perceived effort across later trials. Future work should consider counterbalancing response order to better isolate the unique contribution of each AI response type.

5 Conclusion

This study examined how different styles of generative AI chatbot responses affect students' cognitive load while performing GRE-style verbal reasoning tasks. By integrating physiological data (pupil diameter and fixation duration) with subjective workload ratings, we identified how varying communication strategies of AI assistance influence mental effort during problem-solving. Our findings confirmed that AI assistance improved performance overall, with task accuracy increasing from 30.6% in the Baseline (no AI) condition to 74.0% in the Standard AI condition. However, while performance gains were consistent, the cognitive cost varied notably across different response styles. Physiological measures provided key insights into this variation. Mean pupil diameter was highest in Baseline condition ($M = 3.44 \pm 0.52$ mm), suggesting the highest cognitive load when no assistance was provided. In contrast, all AI-assisted conditions exhibited slightly reduced pupil dilation: Standard (3.41 ± 0.55 mm), Long (3.31 ± 0.54 mm), Unstructured (3.36 ± 0.55 mm), and Ambiguous (3.38 ± 0.56 mm). These decreases imply that AI support can alleviate cognitive load, though the degree of relief is moderated by the clarity and organization of the chatbot's response. Fixation duration patterns offered a complementary perspective. Interestingly, the Standard condition yielded the highest mean fixation duration ($M = 549.31 \pm 207.83$ ms), possibly indicating increased engagement or sustained attention due to coherent and structured AI guidance. Conversely, the "Lengthy" condition resulted in the shortest fixation duration ($M = 428.43 \pm 149.80$ ms), suggesting cognitive disengagement or fatigue from overly verbose explanations. Unstructured (448.07 ± 154.59 ms) and Ambiguous (452.28 ± 161.81 ms) responses also led to relatively shorter fixations compared to the Standard format, reinforcing the notion that disorganized or unclear messaging increases cognitive friction and potentially limits users' ability to process information effectively. Correlational analyses further confirmed a strong relationship between physiological and perceived workload, validating the reliability of eye-tracking as a non-intrusive, real-time indicator of cognitive strain. Specifically, average pupil diameter was positively correlated with subjective mental demand scores ($r = 0.61$, $p < 0.01$), supporting prior literature linking ocular metrics to cognitive stress.

Overall, these results emphasize that not all AI support is cognitively equal. Although all AI conditions reduced pupil dilation relative to baseline, only structured and concise responses maintained high levels of user engagement without inducing additional mental strain. Subjective workload ratings corroborated these findings, with participants reporting higher mental demand and frustration for ambiguous or unstructured outputs. These findings have practical implications for designing conversational AI systems in educational settings. Developers should prioritize clear, structured, and succinct

response styles to minimize unnecessary cognitive load while maintaining user focus. Overly detailed, vague, or unstructured AI explanations may hinder learning by requiring users to compensate cognitively for poor communication design. In future work, dynamic AI systems that adapt their response format based on real-time physiological feedback may offer a promising path forward. This approach could ensure that AI-generated assistance remains cognitively appropriate for individual learners, thereby maximizing both learning effectiveness and user experience. In summary, while AI can enhance task performance, its effectiveness as a learning aid depends critically on how it communicates. Effective AI support must be not only accurate, but also cognitively considerate—facilitating reasoning without overwhelming the learner.

References

Feng, C.M., Botha, E., Pitt, L.: From HAL to GenAI: optimizing chatbot impacts with CARE. Bus. Horiz. **67**(5), 537–548 (2024). https://doi.org/10.1016/j.bushor.2024.04.012

Holmes, W., Bialik, M., Fadel, C.: Artificial Intelligence in Education: Promises and Implications for Teaching and Learning. Center for Curriculum Redesign (2019)

Sweller, J.: Cognitive load during problem solving: effects on learning. Cogn. Sci. **12**(2), 257–285 (1988)

Whitenton, K.: Minimize Cognitive Load to Maximize Usability. Nielsen Norman Group (2013). https://www.nngroup.com/articles/minimize-cognitive-load/

Darejeh, A., Marcusa, N., Mohammadi, G., Sweller, J.: A critical analysis of cognitive load measurement methods for evaluating the usability of different types of interfaces: Guidelines and framework for human-computer interaction (2024). https://doi.org/10.48550/arXiv.2402.11820

Galant-Gołębiewska, M., Zawada, W., Maciejewska, M.: Analysis of Pilot's Cognitive Overload Changes During the Flight (2020). https://doi.org/10.3849/AIMT.01408

Dergaa, I., et al.: From tools to threats: a reflection on the impact of artificial-intelligence chatbots on cognitive health. Front. Psychol. **15**, 1259845 (2024). https://doi.org/10.3389/fpsyg.2024.1259845

Seran, C.E., Tan, M.J.T., Karim, H.A., Al Dahoul, N.: A conceptual exploration of generative AI-induced cognitive dissonance and its emergence in university-level academic writing. Front. Artif. Intell. **8**, Article 1573368 (2025). https://doi.org/10.3389/frai.2025.1573368

Zhai, C., Wibowo, S., Li, L.D.: The effects of over-reliance on AI dialogue systems on students' cognitive abilities: a systematic review. Smart Learn. Environ. **11**, Article 28 (2024). https://doi.org/10.1186/s40561-024-00316-7

Chandler, P., Sweller, J.: Cognitive load theory and the format of instruction. Cogn. Instr. **8**(4), 293–332 (1991). https://doi.org/10.1207/s1532690xci0804_5

Gerlich, M.: AI tools in society: Impacts on cognitive offloading and the future of critical thinking. Societies **15**, Article 6 (2025). https://doi.org/10.3390/soc15010006

Biondi, F., Cacanindin, A., Douglas, C., Cort, J.: Overloaded and at work: investigating the effect of cognitive workload on assembly task performance. Hum. Fact. **63**(5), 813–820 (2020). https://doi.org/10.1177/0018720820929928

Liang, N., et al.: Using eye-tracking to investigate the effects of pre-takeover visual engagement on situation awareness during automated driving. Accid. Anal. Prev. **157**, 106143 (2021)

Tolvanen, O., Elomaa, A.-P., Itkonen, M., Vrzakova, H., Bednarik, R., Huotarinen, A.: Eye-tracking indicators of workload in surgery: a systematic review. J. Invest. Surg. **35**(6), 1340–1349 (2022)

Torres-Salomao, L., Mahfouf, M., El-Samahy, E.: Pupil diameter size marker for incremental mental stress detection. In: 2015 17th International Conference on e-Health Networking, Application & Services (HealthCom) (2015)

Holmqvist, K., Nyström, M., Andersson, R., Dewhurst, R., Jarodzka, H., Van de Weijer, J.: Eye Tracking: A Comprehensive Guide to Methods and Measures. Oxford University Press (2011)

Mayer, R.E.: The Cambridge Handbook of Multimedia Learning. Cambridge University Press (2005)